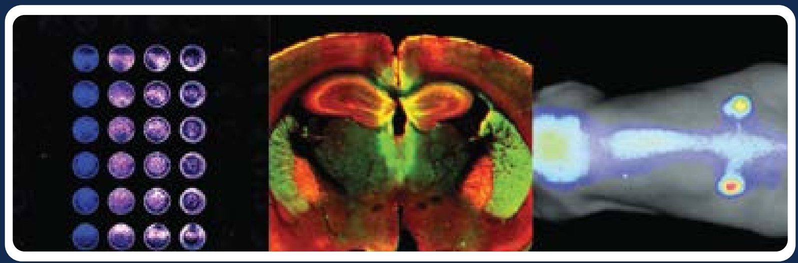
Detailed description of Tissue Section Imaging



Explore a tissue sectioning protocol with LICORbio's Odyssey® Imagers

Tissue section imaging is useful for identifying cellular or tissue localization and expression of biological targets ex vivo. Targets can include proteins, DNA/RNA, and labelled small molecules. There are a variety of methods for target detection, including antibody-based detection, visual stains, and labelled probes. With the appropriate equipment, you can assess both the macro- and micro-locations of your target.

This application outlines the protocol for tissue section imaging using LICORbio's Odyssey® Imagers

Keywords: Tissue section imaging, IHC, Immunostaining, Odyssey,

Introduction

Overview of Immunohistological Staining

Tissue section imaging is vital to advancing scientific understanding across many disciplines. In clinical and research settings, scientists use tissue sections to diagnose illnesses, to assess therapeutic agent colocalization, to understand diseases, and to characterize treatments.

Figure 1 outlines a general protocol for preparing and staining tissue sections. Details of each step are described in the upcoming sections.

As seen in the workflow, several steps will vary with the requirements of the specific assay. The most conventional protocol for tissue section imaging has been described in this application.

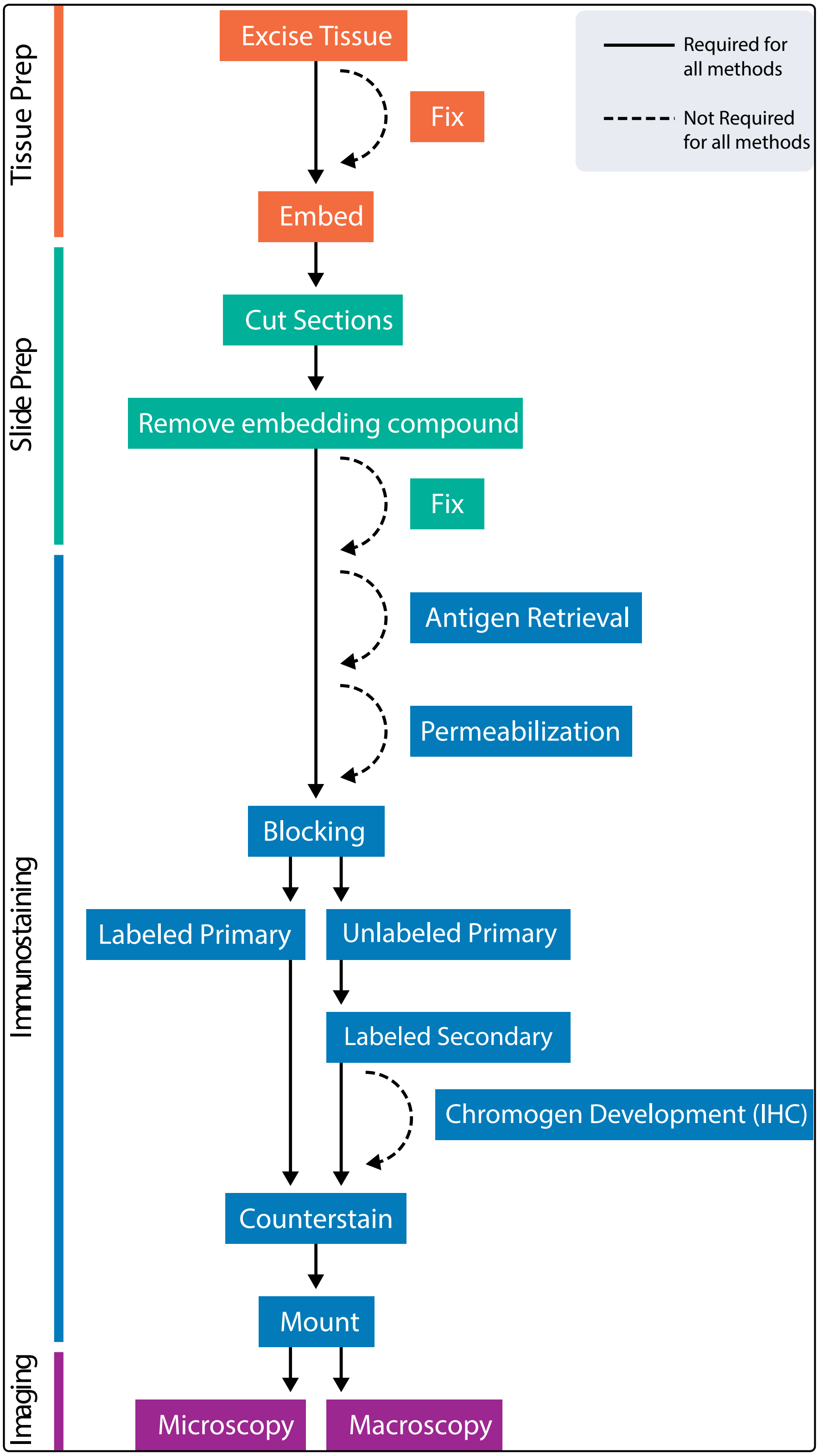


Figure 1: General protocol for preparing and staining tissue sections.

I. Tissue Preparation

1) Fixation

The goal of fixation is to preserve the tissue's cellular structure as closely as possible to its native state.

To lessen risk of sample damage, it is suggested to test several fixative options to determine which fixative preserves both antibody binding and the structural integrity of the sample.

• Fixative Options

There are two common classes of fixatives: cross-linking fixatives and precipitating fixatives.

• Cross-linking fixatives

These fixatives act by creating covalent bonds between macromolecules in the tissue. Crosslinking can mask antigens, so these fixation methods are typically followed by an antigen retrieval step. The most common fixative in histology is formaldehyde gas dissolved in water.

• Precipitating fixatives

Alcohols are commonly used to fix frozen sections and smears. Alcohols permeabilize as they fix and do not induce cross-linking, eliminating the need for later permeabilization and antigen retrieval steps.

• Fixation Factors

While certain structures may be better preserved using specific fixation methods, there are several important factors to consider like pH, osmolarity, duration, tissue thickness, fixative volume and temperature.

2) Embedding

Tissue must be either fixed or frozen immediately after collection to prevent loss of tissue morphology and target protein degradation.

	Paraffin-Embedded	Frozen
Fixation	before embedding	After sectioning if needed
Sectioning Tool	Microtome	Cryotome
Viability	Years at room temperature	~1 year at - 80 °C
Advantages	Preserves cell structures and proteins	Preserves enzyme activity, proteins remain in native state
Disadvantages	Toxic fixatives, proteins are denatured, fixation can shrink tissue and mask epitope	Vulnerable to rapid degradation, ice crystals can disrupt tissue structure

Table 1: Comparison between paraffin and frozen embedding approaches.

Tissues are typically preserved paraffin-embedded or frozen. Table 1 compares these two approaches.

II. Slide Preparation

1) Cut Sections

For microscopy, sections are generally cut to a thickness of 4 to 10 µm so that the section is one cell thick.

Tissues of any thickness can be imaged on an Odyssey® M, Odyssey DLx, or Odyssey XF Imager. Optimal thickness depends on the tissue and embedding method.

A microtome or cryotome is used to cut thin slices of paraffin-embedded tissue or frozen tissue sections respectively.

Once cut, tissue sections are mounted on microscope slides and allowed to dry overnight prior to immunohistochemistry. Alternatively, sections can be transferred to well plates for free-floating section imaging.

2) Remove Embedding Compound

For microscopy, sections are generally cut to a thickness of 4 to 10 µm so that the section is one cell thick.

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3) Free-Floating Sections

In free-floating immunohistochemistry, sections are not mounted until after the immunohistochemistry process is complete. Sections are stained while floating in solution, typically in a large-welled plate. Because the antibody can penetrate the tissue from all sides, thicker sections can be used compared to slide-mounted sections (20 to 50 µm).

III. Immunostaining

1) Perform Antigen Retrieval

Fixation can alter protein biochemistry, causing the epitope of interest to be masked from the primary antibody by cross-linking of amino acids within the epitope or cross-linking with unrelated biomolecules at or near the epitope.

The process of epitope/antigen retrieval refers to unmasking the epitope to restore epitope/antibody binding. It should only be performed if necessary because harsh conditions, including acidic or basic pH and high temperatures, can cause the tissue to dissociate from the slide.

An epitope can be unmasked in several ways.

Heat-induced epitope retrieval (HIER): Epitopes are unmasked by heating the sections in a buffered solution. Citrate and EDTA-based buffers are commonly used. A variety of heating sources can be used, including a microwave, water bath, pressure cooker, or vegetable steamer. Epitope unmasking improves as temperature increases. Water baths and microwaves typically achieve temperatures of 100°C, while pressure cookers can reach up to 120°C.

Proteolytic-induced epitope retrieval (PIER): Enzymes such as trypsin, proteinase k, pronase, or pepsin are used to unmask the antigen.

Room temperature epitope retrieval (RTIER): Acid, typically formic acid (pH 2) or hydrochloric acid (pH 1), is used to reverse cross-linking.

2) Permeabilize

Permeabilization facilitates the entry of antibodies into the cells. The extent of permeabilization required depends on the target.

Permeabilization, like fixation, can affect the morphology of cells, including the antigen of interest.

Detergents commonly used for permeabilization include NP-40, Triton® X-100, Tween® 20, and saponin. After permeabilization, wash the slide in water or PBS.

3) Block

- Create a Hydrophobic Barrier

Prior to adding blocking solution or primary antibodies, draw a barrier on the slide around the tissue section using a hydrophobic pen. The barrier keeps reagents from flowing off the tissue section, allowing application of solutions directly to the slide in smaller amounts.

- Block Endogenous Biotin or Enzyme Activity

Blocking endogenous biotin or enzymatic activity may be needed depending on the detection method. To determine if supplemental blocking of enzymatic activity is needed, add chromogen substrate and/or labelled probe to a control tissue section after antigen retrieval. If endogenous levels of the enzyme produce staining, additional blocking is needed, as described below. This step is typically not needed for fluorescence detection.

- Block endogenous biotin

If you intend to use a secondary antibody conjugated to biotin, block endogenous biotin by incubating the tissue in unlabelled streptavidin or avidin, followed by an excess of unlabelled biotin. This will prevent off-target binding of your labelled streptavidin-conjugated secondary antibody.

- Block endogenous peroxidase

If horseradish peroxidase (HRP) is used for chromogen development, quench endogenous peroxidase by incubating slides in 0.3 to 3.0% (v/v) peroxide in methanol.

- Block endogenous alkaline phosphatase

If the secondary antibody is labelled with alkaline phosphatase (AP), endogenous AP activity can be blocked with the AP inhibitor levamisole at 1 mM concentration.

These blocking steps may interfere with primary antibody binding and can be performed after the primary antibody incubation and wash step if preferred.

• Block Non-Specific Antibody Binding

Target detection can be improved by blocking the non-specific binding of antibody to the tissue section. Intercept® Blocking Buffer (TBS, PBS, or protein-free), BSA, milk, or serum are commonly used to reduce non-specific binding.

Add blocking buffer directly to each tissue section and incubate the slides in a humidified chamber to prevent drying. Each blocking solution will have specific incubation conditions.

4) Incubate With Primary Antibody

Dilute the primary antibody in the same blocking buffer used in the previous step. The addition of detergent helps prevent non-specific interactions. A concentration of 0.1 to 0.2% Tween® 20 is commonly used. Incubate slides in a humidified chamber to keep them from drying out. Most primary antibodies perform well when incubated overnight at 4°C. Some may also perform well at room temperature or 37°C for shorter incubation times.

For multiplexed target detection, be certain each primary antibody is derived from a different host species or subclass and is validated to confirm there is no cross-reactivity. If all antibodies have the same optimal incubation times and temperatures, they can be incubated simultaneously.

After incubating the primary antibody, wash the slides with a buffer containing detergent. Use the same buffer system as in the previous steps. TBS + 0.1% Tween 20 (TBS-T) is commonly used. If the primary antibody is conjugated to a fluorophore, no secondary antibody incubation is needed.

5) Incubate With Secondary Antibody

Dilute the secondary antibody in the same blocking buffer used in previous steps with the addition of detergent. A concentration of 0.1 to 0.2% Tween® 20 is commonly used.

For IRDye® Secondary Antibodies, the recommended starting dilution is 1:200. For VRDye™ Secondary Antibodies, the recommended starting dilution is VRD Rec. Incubate slides in the humidified chamber to keep slides from drying out.

For detection of multiple targets, all secondary antibodies can be incubated simultaneously if they are compatible and will not bind each other.

a. Fluorophore-Conjugated Antibodies

For multiplex detection, make sure fluorophores are compatible and that wavelengths are separated enough to prevent bleedthrough for specific detection of each. For example, the 488, 700, and 800 nm channels on the Odyssey® M Imager are far enough apart for specific detection, while the 488 and 520 nm channels are not. Also, consider that cellular autofluorescence is brightest in the visible wavelengths, so NIR fluorophores will give better sensitivity and reduced background.

b. Enzyme-Conjugated Antibodies

When using secondary antibodies conjugated to an enzyme, such as HRP or AP, you will need to add a substrate. The enzyme breaks down the substrate into a visible precipitate that stains the tissue. The colour of the stain depends on the substrate used. The stain should be developed immediately after secondary antibody incubation and subsequent washes, as enzymes are not stable at room temperature. Stain development must be watched closely, and the reaction must be quenched as soon as staining is satisfactory.

c. Biotin-Conjugated Antibodies

Biotin binds strongly to its ligands avidin and streptavidin. Targets labelled with biotin are often visualized by adding avidin or streptavidin conjugated to a fluorophore or enzyme, like HRP or AP, for chromogenic development, as described above.

6) Counterstain

Slides can be counterstained to visualize overall tissue morphology. Common fluorescent counterstains include nuclear stains, like DAPI, or fluorophore-conjugated phalloidin to stain Factin.

Counterstains are typically performed during or after the secondary antibody incubation.

Visible counterstains for IHC include hematoxylin, nuclear fast red, and methyl green. These can be detected with the RGB channel on the Odyssey® M Imager.

IV. Imaging

1) Mount

Fluorescent Detection

Mounting media can protect against physical damage of the sample and photobleaching of incubation is needed. fluorescent dyes. Some mounting media will solidify the coverslip on the slide, while others will remain liquid.

For liquid mounting media, the edges of the coverslip will need to be sealed with epoxy or clear nail polish. Mounting media is typically added directly to the slide. The coverslip is carefully placed on top to avoid creating air bubbles.

Visible Detection

Although visual stains are not recommended, for visible detection, dehydrate slides through a series of increasing concentrations of alcohols followed by xylenes and allow slides to air dry prior to mounting the coverslip. Clear, adhesive mounting media is typically used.

2) Image

Tissue sections can be imaged using the LICORbio Odyssey® M, Odyssey F, or Odyssey XF Imager.

Ensure slides are completely dry before imaging on an Odyssey® Imager to prevent the mounting media or sealant from damaging the scan surface.

LICORbio Odyssey Imagers include image acquisition and analysis software.

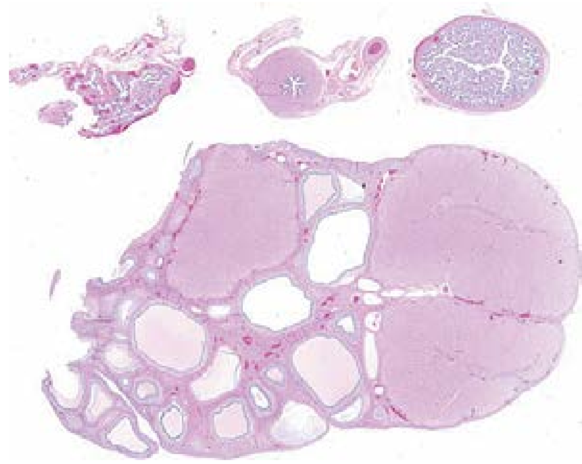

Figure 3: An example tissue section using a pig ovary. Ovaries were deparaffinized, stained using H&E stain, then imaged at 5 µm using the RGB Trans channel of an Odyssey® M Imaging System.

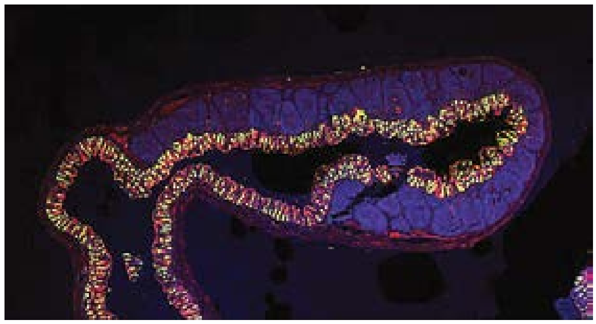

Figure 4: Swine intestine stained for DNA, β-Catenin, and MUC2. The tissue was stained for nuclei (NucSpot 488, blue), β-Catenin (anti- β-Catenin and IRDye® 680RD Goat anti-Mouse Secondary Antibody, red), and MUC2 (anti-MUCS and IRDye® 800CW Goat anti-Rabbit Secondary Antibody, green) and imaged at 5 µm resolution on an Odyssey® M Imaging System.

Conclusion

Odyssey M, Odyssey F, and Odyssey XF Imagers offer macroscopic imaging of tissue sections, allowing for detection of tissue- or region-specific localization and expression of targets. Additionally, these imagers can screen multiple tissue slides simultaneously, offering increased throughput for imaging.

The Odyssey M and Odyssey F Imagers can perform macroscopic imaging of multiple tissue section slides at once, allowing for high-throughput detection of tissue- or region-specific localization and expression of targets. Additionally, Odyssey Imagers provide an industry-leading 6 logs of dynamic range, and the line scanning technology of the Odyssey M delivers exceptionally fast acquisitions of tissue section images.

Instrument	Resolution	Channels
3-Channel Odyssey F	21 – 337 µm	• 700 nm (2 channels) • 800
10-Channel Odyssey F	21 – 337 µm	• 488 nm (4 channels) • 520 nm (3 channels) • 700 nm (2 channels) • 800 nm
Odyssey M	5 – 100 µm	• Onboard Chemi (Optional) • RGB Transillumination (4channels) • RGB Epi-illumination (4channels) • 488 nm (4 channels) • 520 nm (3 channels) • 700 nm (2 channels) • 800 nm

LICORbio's Odyssey Imagers

Odyssey® Family